**A**

Rest-frame Time (yr, z=4.3)

−25 −20 −15 −10 −5 0

F090W–F150W Stacked

$\Delta m_{\rm max} = 0.21 \pm 0.21$

$\chi^2_{\rm red} = 0.35,\ p = 6.25 \times 10^{-1}$

F150W2

$\Delta m_{\rm max} = 0.50 \pm 0.16\ (3.1\sigma)$

$\chi^2_{\rm red} = 3.44,\ p = 1.6 \times 10^{-2}$

F200W

$\Delta m_{\rm max} = 0.16 \pm 0.17\ (0.9\sigma)$

$\chi^2_{\rm red} = 0.33,\ p = 8.1 \times 10^{-1}$

F210M

$\Delta m_{\rm max} = 0.32 \pm 0.18\ (1.8\sigma)$

$\chi^2_{\rm red} = 1.11,\ p = 3.4 \times 10^{-1}$

F277W

$\Delta m_{\rm max} = 0.61 \pm 0.14\ (4.4\sigma)$

$\chi^2_{\rm red} = 7.08,\ p = 9.4 \times 10^{-5}$

F300M

$\Delta m_{\rm max} = 0.69 \pm 0.14\ (5.0\sigma)$

$\chi^2_{\rm red} = 9.23,\ p = 4.2 \times 10^{-6}$

F322W2

$\Delta m_{\rm max} = 0.54 \pm 0.14\ (3.9\sigma)$

$\chi^2_{\rm red} = 5.76,\ p = 6.2 \times 10^{-4}$

F356W

$\Delta m_{\rm max} = 0.52 \pm 0.14\ (3.8\sigma)$

$\chi^2_{\rm red} = 5.67,\ p = 7.0 \times 10^{-4}$

F410M

$\Delta m_{\rm max} = 0.50 \pm 0.14\ (3.7\sigma)$

$\chi^2_{\rm red} = 4.92,\ p = 2.0 \times 10^{-3}$

F444W

$\Delta m_{\rm max} = 0.55 \pm 0.14\ (4.0\sigma)$

$\chi^2_{\rm red} = 5.59,\ p = 7.9 \times 10^{-4}$

−140 −120 −100 −80 −60 −40 −20 0

$t_0 + \Delta t_{\rm obs} + \Delta t_{\rm grav}$ (yr)